

## RUCHA MENE

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### PROFESSIONAL SUMMARY

Software engineering intern with hands-on experience developing, debugging, and supporting production applications across backend APIs, databases, and cloud infrastructure using Python, Java, C++, and JavaScript. Delivered production work spanning OS-level build systems, PKI/security infrastructure, and secure AI chat applications, alongside automation tooling (UiPath, Power Automate, Playwright, Selenium). Strong analytical foundation (500+ LeetCode, Contest Rating 1454). Comfortable managing multiple concurrent deliverables with clear stakeholder communication in high-accountability environments.

### EDUCATION

Vellore Institute of Technology | Bhopal, India

June 2022 – June 2026

B.Tech in Computer Science and Engineering — GPA: 8.4/10

### TECHNICAL SKILLS

**Languages:** Python, Java, C++, JavaScript (ES6+), TypeScript, Go, SQL, Bash/Shell Scripting

**Automation & Testing:** UiPath, Power Automate, Playwright, Selenium, Postman, GitHub Actions CI/CD, Automated Test Suites

**Backend & Frameworks:** Spring Boot, Node.js, Express.js, FastAPI, REST API Design, JWT/Auth, RBAC, Electron (IPC)

**Databases & Analytics:** MySQL, PostgreSQL, MongoDB, Snowflake, Power BI — query optimization, schema design, data validation

**Cloud, OS & DevOps:** AWS (EC2, S3, Lambda), Docker, CI/CD (GitHub Actions), Git/GitHub, Linux/Unix, FreeBSD, React Native (Android & iOS)

**Engineering Practices:** OOP, DSA, System Design, Debugging & Troubleshooting, Production Support, Agile/Scrum, Technical Documentation

### EXPERIENCE

Jai AI HCE Secure | Software Engineer Intern

Feb 2026 – June 2026

Remote, Indore, India

- Built a security-hardened bootable ISO on FreeBSD 14.0 using Poudriere, engineering a 550-package build pipeline and resolving circular dependencies and chroot-level failures, validated through full boot-to-console testing.
- Engineered backend PKI infrastructure (C++, PostgreSQL) across three enterprise services, including a full Oracle-to-PostgreSQL database migration and ASN.1/DER certificate encoding fixes validated against OpenSSL.
- Built the client for a secure AI chatbot platform with end-to-end prompt encryption over a custom transport protocol, multi-model selection, and tool integrations (summarization, presentation generation, insight extraction), alongside a 29-screen React/TypeScript RAG application integrated with live SDK APIs.
- Extended the secure chat and PKI platforms to an 18-screen React Native mobile app (Android & iOS) and built a CLI migration tool (published to npm) with automated rollback and verification across 7 test scenarios, while designing a shared UI/design system reused across multiple applications.

### PROJECTS

#### StockSentry — Financial Data Automation & Analytics Pipeline

**Tech Stack:** Python, Java, SQL, Node.js, AWS (S3, Lambda, EC2), Apache Airflow, Snowflake, Power BI, Docker, GitHub Actions CI/CD

- Automated an end-to-end data ingestion, validation, and transformation pipeline across 50K+ financial records using Apache Airflow, reducing manual processing time by 80%.
- Designed automated data quality rules and Power BI dashboards for pipeline observability, enabling real-time monitoring of data quality metrics and anomaly detection outputs.
- Deployed CI/CD automation via Docker and GitHub Actions, achieving sub-200ms API response times with full end-to-end monitoring and alerting.

#### SafeHer — Backend Application with Production-Grade Support Architecture

**Tech Stack:** Python, Java, Node.js, Express.js, MySQL, MongoDB, AWS (EC2, S3, Lambda), Docker, Playwright, JWT/Auth, REST APIs, CI/CD

- Designed and developed RESTful APIs for real-time event management, applying OOP principles and enforcing data integrity validation across all service boundaries.
- Built automated end-to-end Playwright test suites and troubleshoot production issues under time constraints, maintaining 99% uptime over 6 months with sub-2-second processing.
- Optimized cross-database queries across MySQL and MongoDB, reducing API latency by 40% under concurrent load through schema and index tuning.

#### All About Coding — SQL-Managed Data Pipeline Platform

**Tech Stack:** Python, SQL (MySQL), Node.js, Express.js, React.js, MongoDB, Selenium, Docker, GitHub Actions CI/CD, REST APIs

- Designed normalized SQL schemas and Python ETL pipelines with automated validation rules, reducing data quality issues by 30% and improving content delivery by 35%.
- Implemented automated regression testing using Selenium, integrating test gates into GitHub Actions CI/CD to enforce zero-defect production deployments.
- Built Node.js/Express.js backend integrating third-party content services via REST APIs, with full technical documentation of data contracts and process logic.

### ACHIEVEMENTS

- LeetCode** — 500+ DSA problems solved on LeetCode; Contest Rating 1454 — strong analytical and complex problem-solving skills applicable to debugging and troubleshooting.
- Hackathon Finalist** — Built a financial data pipeline with REST API integration and ML achieving 95% fraud detection accuracy under time-constrained delivery conditions.
- Open Source Contributor** — GirlScript Summer of Code (GSSoC'24); merged 10+ PRs in Python and backend repositories, demonstrating collaborative version control and code review practices.
- Event Lead** — Organized 5+ technical events with 5,000+ participants; coordinated cross-functional teams and maintained documentation standards across all deliverables.

### CERTIFICATIONS

- Oracle Data Platform 2025 — Foundations Associate — Cloud Data Platform & Architecture Certification
- Database Structures and Management with MySQL (Coursera) — SQL, Relational Databases, Query Optimization